

Displacement time graphs

Displacement time graph: gradient of a displacement time graph gives us velocity of the object

Describing displacement time graphs:

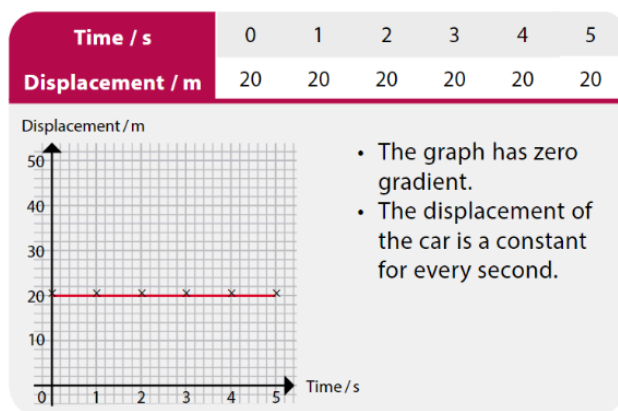
- Velocity
- direction
- reference to starting point

Car moves in opposite direction from which is started (negative displacement) when displacement is below the X axis

when graph is decreasing, but still above x axis $\Rightarrow$ car moves back to its original point $\Rightarrow$ from which it started

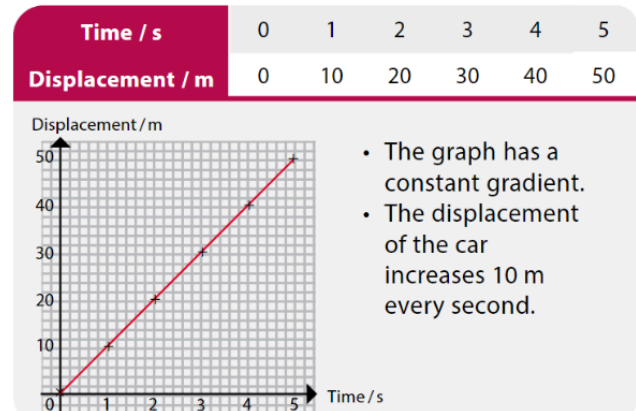
When this happens: state: the care is still in positive direction form starting point, but it is moving back towards starting point

Car at rest



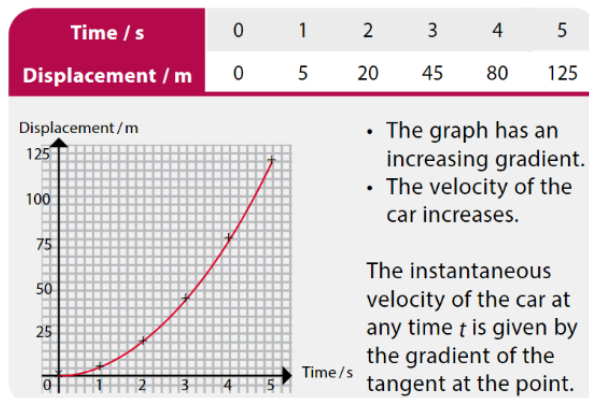
- Graph has 0 gradient $\rightarrow$ thus velocity is 0.
- Displacement is constant for every second.
- Car is at rest

Car travelling at a uniform velocity of 10 m/s



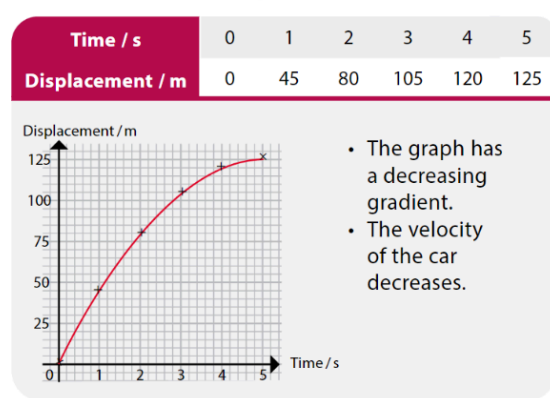
- Graph has constant gradient - uniform velocity (positive velocity) $\rightarrow$ away from reference point*
- Displacement of cars increases 10m every second.
- Moves in a positive direction from the starting point. (check)

Car travelling with increasing velocity (non-uniform velocity)



- Graph has an increasing gradient
- Velocity of the car is increasing (non-uniform)
- Moves in a positive direction from the starting point.

Car travelling with decreasing velocity (non-uniform velocity)



- Graph has decreasing gradient
- Velocity of the car decreases
- Moves in a positive direction from the starting point.